What Households Borrow For: The Effects of Debt Composition on Business and Income Outcomes

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Abstract

The literature on household borrowing primarily focuses on the consumption behavior of highly indebted households, along with the economic consequences caused by excessive household leverage.

Using the Household Finance and Consumption Survey (HFCS), this paper examines the underlying reasons for household borrowing and shows that the economic effects of household debt depend critically on its allocation across purposes, rather than on its aggregate level alone. By employing linear and conditional fixed-effects models across five European countries, the analysis provides novel empirical evidence on the composition of household debt and its effects on households' business and income outcomes.

The results reveal that households rely on external credit to fund secondary business activities beyond their primary wealth source, while borrowing for educational purposes is significantly associated with self-employment and higher employee income. In addition, households compensate for large expenditures such as purchasing the main residence by increasing rental income, although this effect dissipates for households able to borrow larger amounts. Overall, access to credit has stronger effects than the amount borrowed, suggesting that improving credit availability for productive uses may support household-level financial development and broader economic activity.

Keywords: Household Finance, Borrowing, Household Assets, Investment Decisions, Mortgages

JEL Classification: D14 , G51

1 Introduction

In the aftermath of the global financial crisis of 2007-2008, scholarly attention turned to household debt and the risks associated with excessive lending for the economy. For instance, Jordà et al. (2013) find that business cycles characterized by intensive credit expansions preceded steep downturns and slow recoveries. Similarly, Schularick and Taylor (2012) show that credit growth is a strong predictor of financial crises, arguing that modern economic recessions are essentially "credit booms gone wrong".

While uncontrolled lending and weak screening protocols represent an undeniable threat to financial stability, other authors claim that household leverage alone cannot explain the severity of the global recession. For example, Jones et al. (2022) demonstrate that household deleveraging explains only a small fraction of the aggregate employment decline in 2007–2010. Likewise, Gorton and Ordoñez (2020) argue that it is not credit growth alone that predicts crises, but rather its combination with deteriorating productivity. This argument suggests that the resilience of the credit market depends critically on the quality of the investments financed, which do not necessarily result in financial crises. Rather, a sustainable funding structure and solid underwriting standards can well support even instances of high aggregate debt (André, 2016).

Nevertheless, the prevailing academic narrative surrounding household borrowing primarily underlines its macroeconomic risks, emphasizing in particular the spending cuts of highly levered households associated with negative exogenous shocks, which may further amplify economic downturns (Dynan and Edelberg, 2013; Mian et al., 2013; Zabai, 2017). While these studies provide welcome and relevant evidence on the negative consequences of unchecked credit supply, less attention has been devoted to the micro-level allocation of borrowed funds. This distinction is crucial, as different uses of credit may have markedly different implications for both household-level outcomes and aggregate economic dynamics.

Using the longitudinal component of the ECB's Household Finance and Consumption Survey (HFCS), this paper revisits the narrative associating household borrowing with consumption-led uses of leverage, and argues that households borrow to improve their finances by investing in revenue-generating activities. In doing so, the paper argues that the economic effects of household debt depend critically on how credit is allocated across purposes, rather than on its aggregate level alone. This logic follows the "borrow to invest motive" developed by Crossley et al. (2024), where the authors show that, once credit constraints are relaxed through house price increases, leveraged households borrow to invest in additional housing rather than to increase consumption.

This paper exploits the detailed information collected in the HFCS about households' borrowing behavior to identify wealth-generating mechanisms across five European countries − namely, Belgium, Cyprus, Germany, Italy, and Spain − in a period between 2010 and 2020. In particular, the HFCS questionnaire asks if and how the respondents accessed a loan at the time of the interview, and for what purpose. These responses provide a clean way of disentangling productive uses of financial leverage – that is, investments with the potential to generate additional wealth in the future – from purely consumptive debt. This design represents a key distinction from other works analyzing household borrowing, where the latter is either evaluated in its aggregate form at the country level (Mian et al., 2017; Lombardi et al., 2017; Berisha and

Meszaros, 2018), or limited to mortgage debt (Corradin and Popov, 2015; Di Maggio et al., 2017; Bracke et al., 2017).

Using linear and conditional fixed-effects regressions, this paper examines both the extensive and intensive margin effect of purpose-specific borrowing on household business and income outcomes, revealing novel degrees of granularity both in terms of investments and financing channels. To mitigate simultaneity concerns and reverse causality, the analysis uses a rich set of controls and lagged values of the explanatory variables, thus following established approaches in the household finance literature (Dynan, 2012; Du Caju et al., 2023).

The results confirm that the economic implications of household debt strongly depend on its intended purpose. Specifically, households collateralize secondary properties to purchase additional real estate for business activities. In addition, they rely on a range of financing instruments to renovate or refurbish the property used for business, which is in turn associated with significant increases in value. Furthermore, borrowing for business purposes and for vehicle purchases are associated with non-self employment business wealth, suggesting that loans fund secondary activities beyond the primary source of household wealth. By contrast, loans taken for education correspond to a higher probability of having self-employment wealth and employee income, reflecting the relevant role of human capital in employment opportunities. Finally, households cushion the impact of housing purchases by turning to rental income generation, although this effect does not hold for households able to access large credit amounts.

This study makes three important contributions to the literature. First, it demonstrates that the composition of household debt is central to understanding its economic effects. By examining each purpose behind a household's decision to take on debt, the analysis documents substantial heterogeneity in how leverage is allocated across consumption, real estate investment, entrepreneurial activity, and human capital accumulation. This distinction provides a more precise assessment of how debt contributes to wealth accumulation at the household level, which is typically obscured when debt is measured at the aggregate level or proxied through consumption metrics.

Second, the paper contributes to the literature on household leverage and entrepreneurship by identifying multiple financing channels beyond home equity extraction. While prior work emphasizes the role of housing collateral in facilitating entrepreneurial entry (Schmalz et al., 2013; Adelino et al., 2015; Jensen et al., 2022), this study considers collateralized and non-collateralized borrowing instruments and links each financing channel to specific investment purposes and wealth outcomes. In doing so, the analysis provides a more comprehensive view of how households mobilize credit to support income-generating activities, expanding the entrepreneurial scope of household leverage.

Third, the paper contributes to the growing literature using the HFCS as reference data (Bover et al., 2016; Du Caju et al., 2023). By constructing a balanced panel dataset that includes only households interviewed in all survey waves, this work represents one of the first attempts at analyzing the longitudinal component of the data. While the absence of panel weights limits external generalizability, the within-household design provides new evidence on the dynamic relationship between borrowing decisions and subsequent wealth changes.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature and develops the hypotheses. Section 3 describes the data and variable construction. Section 4 outlines the empirical strategy. Section 5 presents the results, and Section 6 concludes.

2 Literature Review

A large literature studies the determinants and macroeconomic consequences of household debt. One strand attributes the rapid expansion of credit prior to the global financial crisis primarily to supply-side factors. Mian and Sufi (2009) show that mortgage growth across U.S. counties was driven mainly by an outward shift in credit supply rather than improved borrower fundamentals. Keys et al. (2010) document that securitization weakened lenders' screening incentives, increasing default risk, while Justiniano et al. (2019) emphasize financial liberalization and looser borrowing constraints as central drivers of the housing boom. More broadly, house price dynamics and monetary conditions have been identified as key long-run determinants of household leverage (Jordà et al., 2015; Moore and Stockhammer, 2018b). Cross-country evidence further highlights the role of legal institutions, bankruptcy regimes, and financial structures in shaping borrowing patterns (Bover et al., 2016; Coletta et al., 2019).

A second strand focuses on the macroeconomic consequences of rising household indebtedness. Jordà et al. (2016) document the central role of mortgage credit in financial instability, and Mian and Sufi (2018a) argue that recessions are amplified by deleveraging among highly indebted households. Numerous studies show that debt can amplify downturns by concentrating spending cuts among high marginal-propensity-to-consume households (Dynan, 2012; Mian et al., 2013; Zabai, 2017; Du Caju et al., 2023). Related work documents heterogeneous consumption responses to mortgage refinancing and interest rate changes (Di Maggio et al., 2017; Cloyne et al., 2020).

While this literature provides compelling evidence on the systemic risks associated with excessive leverage, it typically treats household debt as a relatively homogeneous aggregate. Less attention has been devoted to the microeconomic allocation of borrowed funds and to whether different forms of credit serve distinct economic functions. Some studies suggest that the relationship between debt and spending is more nuanced than aggregate measures imply. Andersen et al. (2016) argue that high debt levels may reflect intertemporal spending normalization rather than persistent vulnerability. Baker (2018) show that consumption sensitivity depends critically on liquidity and borrowing constraints, while Mason (2018) document that only a small share of borrowing is directly used for consumption expenditures. Together, these findings suggest that the economic consequences of debt may depend on how credit is allocated across uses.

A growing body of research points to specific channels through which household borrowing may support income-generating activities. One prominent mechanism operates through housing collateral. Access to home equity relaxes borrowing constraints and increases entrepreneurial entry and business investment (Schmalz et al., 2013; Adelino et al., 2015; Corradin and Popov, 2015; Jensen et al., 2022). This evidence implies that households with access to financing invest in productive endeavors such as self-employment activities rather than merely to finance consumption.

Hypothesis 1: Borrowing, particularly when supported by collateral, is positively associated with business-related activities.

Beyond housing collateral, borrowing may also enhance labor market opportunities. Van Doornik et al. (2024) show that access to credit for transportation investments expands commuting possibilities and raises wages and formal employment. Similarly, vehicle ownership reduces spatial mismatch and increases employment prospects (Baum, 2009), while low-cost financing has enabled participation in gig economy activities (Buchak, 2020). These studies suggest that loans for transportation may generate income gains by expanding access to labor markets and business opportunities.

Hypothesis 2: Borrowing to finance transportation is positively associated with additional income-generating activities.

Real estate investments represent another potentially productive use of leverage. Homeownership may contribute to wealth accumulation through capital gains, forced savings, tax advantages, and rental income (Goodman and Mayer, 2018; Jordà et al., 2019; Demers and Eisfeldt, 2021). The expansion of buy-to-let activity and short-term rental markets further indicates that households may actively use debt to generate rental income streams (Bø, 2026).

Hypothesis 3: Borrowing to finance housing purchases is positively associated with rental income generation.

Finally, borrowing for education may increase human capital and long-term earnings. Black et al. (2023) show that expanded student loan access raises degree completion and lifetime income, while Avery and Turner (2012) document heterogeneous borrowing patterns linked to expected returns. These findings underscore that educational debt may represent an investment in future income rather than a purely consumption-oriented liability.

Hypothesis 4: Borrowing for educational purposes is positively associated with higher employment outcomes and earnings over time.

Taken together, this literature suggests that the effects of household debt depend critically on its purpose and allocation. However, systematic evidence distinguishing borrowing by purpose within a longitudinal European household panel remains scarce. This paper contributes to filling this gap by directly linking purpose-specific borrowing to subsequent business and income outcomes. By testing the related hypotheses, the paper evaluates whether and how different components of household debt contribute to wealth and income dynamics.

3 Data and Descriptive Statistics

3.1 Data Source

The dataset used for this analysis is the Household Finance and Consumption Survey (HFCS)[1], a recurring survey conducted approximately every three years that examines the consumption behavior and financial practices of European households. A key feature of the HFCS for this study is that it records not only households' borrowing decisions, but also the stated purpose of each loan and the type of financing instrument used. To date, four survey waves have been conducted by the central banks of participating countries. For most countries, fieldwork for the first wave took place between 2010 and 2011 (2011 wave), between 2013 and the first half of 2015 for the second wave (2014 wave), and in 2017 for the third wave (2017 wave). The fourth wave, originally planned for 2020, was conducted throughout 2020 and 2021 due to disruptions caused by the COVID-19 pandemic (2020 wave). To ensure comparability across waves, all nominal variables are adjusted for inflation using the Harmonised Index of Consumer Prices (HICP).[2]

An important aspect of the dataset is the presence of repeated households, that is, households interviewed in all survey waves. The resulting balanced panel includes approximately 16,500 household-wave observations across five euro area countries: Germany (1,216 households), Italy (1,129 households), Spain (809 households), Cyprus (623 households), and Belgium (335 households). This longitudinal structure enables the analysis of within-household changes over time, which forms the empirical basis of the study. While survey weights are provided for cross-sectional analysis to ensure representativeness of the target population, no corresponding panel weights are available. Accordingly, the empirical analysis prioritizes internal validity through within-household variation using fixed-effects models, rather than external generalizability.

To address missing data and item non-response, the HFCS provides five multiply imputed datasets based on observable characteristics. Following Rubin (1987), all estimates are computed separately for each imputed dataset and combined using Rubin's rules. The construction of key variables is described in detail in the following subsection.

3.2 Descriptive statistics

A list of the main variables used in the analysis is reported in Table 1. In particular, panel A describes the dependent variables examined across various empirical specifications. These are a mix of business- and income-related outcomes. For instance, *Has properties for business* is a dummy indicating whether the household owns any real estate, such as houses, apartments, garages, offices, hotels, farms, and land used for business activities. The corresponding value of the estate is captured by *Value of properties for business*. Similarly, *Has SE business* is a binary indicator reflecting

[1] European Central Bank: Household Finance and Consumption Survey (HFCS). https://www.ecb.europa.eu/stats/ecb_surveys/hfcs/html/index.en.html. Accessed 02 March 2025

[2] The values of assets, debt, income and consumption have been adjusted by multiplying earlier-wave figures by the ratio of average price levels in the corresponding reference years.

whether the household actively run all or part of a self-employment (SE) activity, and *Has non-SE business* refers to a household being a private investor or silent partner of a non-self employment business. In addition, *Value of non-SE business* captures the monetary amount of the household's share in the non-self employment business, while *Has income from non-SE business* is a dummy revealing whether the household received any income from these private activities in the last calendar year. These variables capture household participation in entrepreneurial activities and the corresponding value of business-related assets and income.

Moreover, four dependent variables refer to earnings outcomes. *Has rental income* is a binary indicator for households receiving any income from renting real estate in the last 12 months, and *Value of rental income* captures the corresponding monetary amount. Likewise, *Has employee income* indicates whether the household received any sort of employee income during the last calendar year, and *Value of employee income* reflects its worth.

As mentioned earlier in the text, one of the novelties introduced by this study is the focus on the underlying motivations behind household borrowing. The HFCS collects eight possible reasons for taking out loans: (1) to purchase or construct the Household Main Residence (HMR), (2) to purchase other real estate, (3) to refurbish or renovate the residence, (4) to buy a vehicle or other means of transport, (5) to finance a business or professional activity, (6) to consolidate or refinance debts, (7) for education purposes, and (8) to cover living expenses or other purchases. The frequency of each loan is described in panel B, where around one fifth of households reports having a loan taken to purchase or construct the HMR at any given survey wave (Mean = 0.18). In contrast, the occurrence of the other purposes is lower, consistent with other studies finding that housing debt is the most common form of household borrowing (Causa et al., 2019). While the HFCS allows households to report multiple loan purposes simultaneously, the analysis focuses on single-purpose loans to ensure a clear mapping between borrowing motives and subsequent outcomes, thereby reducing ambiguity in the interpretation of estimated effects.

A similar strategy is implemented for the variables presented in panel C, which describes the euro amount still owed on the purpose-specific loan at the time of the interview. The availability of both the binary indicator and the monetary value of each loan allows the analysis of both the extensive margin (whether a household borrows for a given purpose) and the intensive margin (the amount borrowed for that purpose). This ensures robustness of the results and adds transparency and granularity to the findings.

In addition, panel D shows the summary statistics concerning the instruments used by households to access credit. The HFCS identifies four financing channels available to households, each represented as a binary indicator: (1) mortgages or loans using the HMR as collateral, (2) mortgages or loans using other property as collateral, (3) non-collateralized loans, and (4) private loans. While the latter is the least common borrowing type, with just 2% of households reporting having loans from relatives or friends that are expected to be repaid, mortgages using the HMR as collateral is the most frequent one, reflecting the central role of housing collateral in household borrowing (Adelino et al., 2015; Sodini et al., 2023).

Table 1 Summary statistics

Variable	Mean	SE	Min	Max	N
Panel A. Dependent variables					
Has properties for business	0.04	0.0015	0	1	16,448
Value of properties for business	14,778.20	1,940.2360	0	17,000,000	16,448
Has SE business	0.16	0.0028	0	1	16,448
Has non-SE business	0.03	0.0013	0	1	16,448
Has income from non-SE business	0.03	0.0014	0	1	14,190
Value of non-SE business	106,090.90	33,759.4600	0	350,000,000	16,448
Has rental income	0.18	0.0030	0	1	16,448
Value of rental income	3,030.52	125.6956	0	650,000	16,448
Has employee income	0.57	0.0039	0	1	16,448
Value of employee income	28,879.31	671.6569	0	7,276,530	16,448
Panel B. Loan purpose (Extensive margin)					
To buy the HMR	0.18	0.0030	0	1	16,448
To buy other real estate	0.04	0.0016	0	1	16,448
Property renovation	0.04	0.0016	0	1	16,448
Vehicle	0.06	0.0018	0	1	16,448
Business activity	0.02	0.0011	0	1	16,448
Refinancing debt	0.01	0.0007	0	1	16,448
Education	0.01	0.0008	0	1	16,448
Consumption	0.03	0.0013	0	1	16,448
Panel C. Outstanding balances (Intensive margin)					
Outstanding HMR loans	7,143.84	576.6548	0	4,195,000	16,448
Outstanding other real estate loans	6,164.54	472.7645	0	2,816,000	16,448
Outstanding renovation loans	1,628.66	114.0888	0	580,000	16,448
Outstanding vehicle loans	526.01	25.0035	0	111,000	16,448
Outstanding business loans	3,560.17	1,575.0490	0	23,180,000	16,448
Outstanding refinancing loans	371.07	539.0930	0	3,000,000	16,448
Outstanding education loans	276.86	37.5743	0	240,000	16,448
Outstanding consumption loans	295.53	47.8762	0	490,850	16,448
Panel D. Financing					
HMR mortgage	0.19	0.0031	0	1	16,447
Other property mortgage	0.07	0.0021	0	1	16,436
Non-collateralized loan	0.13	0.0026	0	1	16,410
Private loan	0.02	0.0012	0	1	12,240
Panel E. Controls					
Total household income	35,156.55	561.7567	-965,000	5,032,354	16,448
Optimistic	0.13	0.0027	0	1	16,118
Wealth quintile	3.63	0.0110	1	5	16,448
Household size	2.61	0.0101	1	10	16,448
Age	59.02	0.1104	19	100	16,448
Female	0.30	0.0037	0	1	16,252
Married	0.70	0.0036	0	1	16,224
Employed	0.54	0.0039	0	1	16,438
Tertiary education	0.39	0.0038	0	1	16,181

Notes: Summary statistics and standard errors are calculated using multiple imputations across five complete datasets. Variable definitions and survey questions are reported in Table A.1 and Table A.2.

Finally, panel E describes the rich set of covariates included in the empirical specifications. In particular, *Total household income* represents the gross annual aggregate household income, and controls for differential access to credit across income levels. Similarly, *Wealth quintile* is a categorical variable of 5 values, with 5 representing the wealthiest quintile and 1 the least wealthy, and it controls for initial wealth levels and differing saving behaviors. Furthermore, *Optimistic* is a binary indicator equal to 1 if the household expects its total income to increase more than prices over the next year, while it takes value 0 if the household expects a future income level about the same as prices or lower than prices. This variable reflects households' general attitude towards their future finances, which may affect their risk propensity in taking out loans, and becomes particularly relevant for the earnings models.

Additional demographic controls traditionally found in the wealth and income literature are also included. Following the UNECE (2011) guidelines, these are defined according to the household reference person's responses, and may therefore vary over time[3]. Importantly, to reduce noise and sparse within-household variation in the estimations, *Married*, *Employed*, and *Tertiary education* are binary indicators constructed from categorical variables of the household reference person. In particular, *Married* equals 1 for individuals in a registered marriage and 0 otherwise; *Employed* equals 1 for employees or self-employed individuals and 0 for all other labour statuses; and *Tertiary education* equals 1 for individuals with tertiary qualifications and 0 for lower education levels. A detailed description of the survey questions and variable definitions is provided in Table A.1 and Table A.2 in A.

Lastly, the conditional distribution of purpose-specific loans across the various financing instruments, and their respective mean outstanding amounts, are presented in Table 2. The left panel shows the shares attributable to each financing instrument and loan purpose combination, conditional on having the corresponding loan. Similarly, the right panel reports the mean conditional outstanding amount across each financing-purpose pair.

As observed in Table 1, main residence purchases constitute the most frequent motivation to take out a loan, with nearly 3,000 observations. Their financing is almost exclusively granted through real estate collateral (HMR or other property (OTP)), while the share of non-collateralized loans (NCLs) and private loans (PLs) is negligent.

Table 2 Financing mix and outstanding balances by loan purpose (conditional on having the loan)

		Financing instrument							
		Financing shares (%)				Mean outstanding balance (EUR)			
Loan purpose	Obs.	HMR	OTP	NCL	PL	HMR	OTP	NCL	PL
HMR loan	2,963	85.4	16.6	2.7	1.1	16,547	21,086	1,507	476
Other real estate loan	675	15.2	72.9	13.0	4.0	17,052	119,066	10,998	2,086
Renovation loan	688	50.5	11.3	36.0	6.2	23,528	8,941	5,404	895
Vehicle loan	912	0.5	1.0	94.0	7.2	170	144	8,798	354
Business loan	299	17.3	22.3	57.1	15.9	19,693	120,154	46,339	8,396
Refinancing loan	119	11.7	6.7	74.0	8.7	10,155	7,165	31,835	1,894
Education loan	184	18.9	7.6	68.8	8.0	11,593	4,067	7,771	1,159
Consumption loan	485	2.5	3.3	86.8	13.1	1,634	513	7,226	612

Notes: Entries are computed using multiple imputation. Financing shares (left panel) are conditional means of financing indicators, conditional on having the corresponding loan purpose. Shares may sum to more than 100 percent because households can use multiple instruments for the same loan purpose. Mean outstanding balances (right panel) are conditional means of outstanding amounts by financing type, also conditional on having the corresponding loan purpose. "Obs." reports the number of observations used for each purpose-specific conditioning sample.

[3]The reference person is uniquely determined by applying the following sequential criteria: first, a partner in a de facto or registered marriage with dependent children; then, a partner in such a union without dependent children; next, a lone parent with children; if none of these apply, the person with the highest income; and finally, the oldest individual in the household.

A similar situation applies to loans taken to purchase other real estate, where the bulk of the financing comes from collateralizing OTP (72.9%), although the share of NCLs increases to 13%. By contrast, a mixed set of instruments is used for loans taken to renovate the residence – albeit collateralizing the HMR still constitutes half of the financing – and for business loans. For the latter, NCLs are the preferred instrument of credit, although the share of other financing resources is more evenly distributed compared to other types of loans.

Finally, borrowing to purchase vehicles, to refinance debt, to improve human capital through education, and to cover living expenses all reveal a penchant for using NCLs as financing instrument. These patterns highlight the importance of considering multiple financing channels when analyzing household borrowing behavior. The empirical framework therefore distinguishes borrowing along three dimensions: purpose, financing instrument, and borrowing intensity.

4 Model Specification

The methodology used for the analysis is illustrated in Eq. 1:

$$\tilde{Y}_{it} = \alpha_i + \beta \,\mathrm{Loan}^{(p)}_{i,t-1} + \mathbf{X}'_{it}\gamma + \delta_t + \lambda_t + \varepsilon_{it} \tag{1}$$

where $\tilde{Y}_{it}$ is one of the dependent variables introduced in panel A of Table 1. The specification exploits within-household variation over time, effectively controlling for all time-invariant household characteristics that may jointly affect borrowing decisions and economic outcomes. When $\tilde{Y}_{it}$ is binary, equation (1) is estimated as a linear probability model with fixed-effects α_i. For continuous outcomes, it corresponds to a linear fixed-effects (within) model.

The key explanatory variable $\mathrm{Loan}^{(p)}_{i,t-1}$ is the lagged value of the loan variable corresponding to purpose–financing combination p. The reason behind using its lagged value is twofold: (1) it allows time for the effects of borrowing to materialize in subsequent outcomes; (2) it mitigates concerns related to the correlation between credit access and economic variables. By regressing borrowing decisions at time $t-1$ to outcomes at time t, the specification reduces concerns related to simultaneity and reverse causality (Dynan, 2012; Du Caju et al., 2023).

In addition, for the specifications using continuous dependent variables, the variable is differenced with its value in the previous wave. This strategy is adopted to further reduce the influence of persistent levels of the outcome variable, increasing consistency and robustness of the results. The control vector $\mathbf{X}'_{it}$ includes the initial levels of wealth and income, along with the other demographic variables described in panel E of Table 1. Furthermore, δ_t are time fixed effects, controlling for macroeconomic shocks common to all households, while λ_t control for all unobserved, time-invariant heterogeneity across countries that might affect the outcome variable. Finally, ε_{it} denotes the residual.

Following Pence (2006), the monetary variables are transformed using the Inverse Hyperbolic Sine (IHS) function to retain zero and negative values included in the dataset. Unlike the logarithmic transformation, the IHS accommodates zero and negative values without requiring arbitrary adjustments, and has become standard practice in the wealth literature (Causa et al., 2019). For large values, the IHS behaves similarly to the logarithmic transformation, allowing coefficients to be interpreted approximately as elasticities, while remaining closer to a linear transformation for small values (Burbidge et al., 1988).

Finally, regarding the specifications having a binary indicator as dependent variable, the analysis estimates conditional fixed-effects logistic models to provide additional robustness checks, as shown in Eq. 2 below:

$$\mathbb{P}\left(Y_{it}^{(b)} = 1 \mid \mathbf{X}_{it}, \alpha_i\right) = \Lambda\Big(\alpha_i + \beta\,\mathrm{Loan}_{i,t-1}^{(p)} + \mathbf{X}_{it}'\gamma + \delta_t + \lambda_t\Big)\,. \tag{2}$$

where the parameters are the same as above. A key feature of these regressions is that the estimation relies exclusively on within-household variation in the dependent variable, while non-varying observations are dropped. As a result, the number of observations reduces significantly, and the estimates are less precise than in a linear probability model. Therefore, the latter is used as reference specification in the main text, while the results from the conditional fixed-effects logistic models are provided in C. Importantly, the results are consistent in sign and statistical significance across specifications.

5 Estimation Results

The results are presented as follows. Subsection 5.1 examines the relationship between borrowing for real estate purposes and business-related outcomes. Subsection 5.2 focuses on the association between purpose-specific borrowing and entrepreneurial activity, distinguishing between self-employment (SE) and non-SE business outcomes. Subsection 5.3 analyzes the link between borrowing and rental income generation, while Subsection 5.4 investigates the relationship between education-related borrowing and employee income outcomes.

5.1 Properties for business

The first set of models evaluates whether households use leveraged real estate investments for business activities. The results provided below focus on the key explanatory variables, while the tables with full controls can be found in B. The dependent variable analyzed in Table 3 is the binary *Has properties for business* introduced in Section 3.2, which captures ownership of real estate used for business activities. To evaluate the hypothesis that households also use securitized borrowing for entrepreneurial activities rather than consumption alone, the explanatory variables of interest are the lagged value of loans collateralized by other properties used to purchase additional real estate (columns 1 and 2), and the lagged value of loans taken using the HMR as collateral to renovate or refurbish the property (columns 3 and 4). The results report both the extensive margin (Dummy) and the intensive margin (Debt Level) effects of the loans.

Table 3 Collateral Channels, Loan Purpose, and the Probability of Holding Real Estate for Business

	DV: Holding real estate for business			
	Purpose: Buy other properties		Purpose: Renovation	
Financing channel	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Other property$_{t-1}$	0.033**	0.003**		
	(0.015)	(0.001)		
Main residence$_{t-1}$			0.025*	0.002*
			(0.014)	(0.001)
Observations	12,286	12,298	12,297	12,298
Controls	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: Entries are coefficients from multiple-imputation linear probability models with household fixed effects. The dependent variable is a binary indicator equal to one if the household holds real estate for business purposes. Columns are grouped by the *purpose of the collateralized loan.* "Dummy" indicates whether the household holds a collateralized loan for the stated purpose (extensive margin). "Debt Level" is the inverse hyperbolic sine (IHS) transformation of the outstanding balance of the collateralized loan for the stated purpose (intensive margin). "Other properties" refers to loans collateralized by other real estate; "Main residence" refers to loans collateralized by the HMR. All main regressors are lagged by one survey wave. Coefficients can be interpreted as changes in probability. Standard errors clustered at the household level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column (1) shows that having a mortgage to purchase additional properties at $t-1$ increases the probability of holding real estate for business activities in the next period by 3.3 percentage points. In addition, a 10% increase in the outstanding loan amount is associated with approximately a 0.03 percentage point increase in probability (column 2). These results suggest that borrowing for real estate acquisition is positively associated with subsequent ownership of business-related real estate.

While the magnitudes are modest, the robustness checks using conditional fixed-effects logit models suggest economically meaningful effects. In fact, analyzing the same effect exclusively among households switching the outcome shows that the odds of holding real estate for business are approximately 98% higher (column 1, Table C.1, C), consistent with the linear probability estimates.

Similarly, the coefficient in column (3) indicates that households' decision to collateralize the HMR for renovation purposes in period $t-1$ is associated with a 2.5 percentage point higher probability of owning real estate for business in the next period, while the intensive margin effect is small in magnitude ($\beta = 0.002$). These findings highlight the relevant role of collateral availability in supporting entrepreneurial activities, and the importance of considering a wider range of purposes of household borrowing beyond consumption.

Moreover, Table 4 examines the effect of these renovation loans on the value change of the property used for business. For columns (2) and (4), since the monetary amounts of both property and loans are IHS-transformed variables, the specifications behave similarly to a log-log model for large values, allowing coefficients to be interpreted approximately as elasticities, while remaining closer to a linear relationship for small values (Burbidge et al., 1988). By contrast, to compute proportional effects when the independent variable is a dummy (columns 1 and 3), I apply the expression proposed by Bellemare and Wichman (2020):

$$\bar{R} \approx e^{\hat{\beta} - 0.5\,\widehat{\mathrm{Var}}(\hat{\beta})}, \tag{3}$$

where $\bar{R}$ represents the ratio of conditional means. Accordingly, the coefficient in column (1) reveals that, on average, having a mortgage using the HMR as collateral for renovation purposes increases the value change of the real estate used for business activities by approximately 55%. Similarly, non-collateralized borrowing to renovate the property is associated with an increase in the value change of the business estate by approximately 50%. The incremental impact of the outstanding loan amount is also positive and statistically significant, hovering around 4-5% for both financing instruments.

Overall, these results show that households use housing collateral to support real estate investments related to business activities, and indicate that the decision to access credit is more strongly associated with entrepreneurial outcomes than the amount borrowed, suggesting that credit access itself may be a key determinant.

Table 4 Renovation Borrowing and the Change in the Value of Real Estate for Business

	DV: Δ IHS value of real estate for business			
	Purpose: Renovation			
Financing channel	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Main residence$_{t-1}$	0.469**	0.041**		
	(0.232)	(0.020)		
Non-collateralized loan$_{t-1}$			0.439*	0.050**
			(0.244)	(0.024)
Observations	12,297	12,298	12,265	12,298
Controls	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: Dependent variable is defined as the change in the inverse hyperbolic sine (IHS) value of other real estate property used for business activities between survey waves. Entries are coefficients from multiple-imputation fixed-effects (within) regressions with standard errors clustered at the household level in parentheses. "Dummy" indicates the presence of a renovation loan (extensive margin). "Debt Level" is the IHS-transformed outstanding renovation debt (intensive margin). All main regressors are lagged by one survey wave. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Private business wealth

This section further explores the purpose-specific effects of household borrowing on business-related outcomes by distinguishing between self-employment (SE) and non-SE business wealth. The former implies that the household has an active role in running the business, while in the latter the household acts as an investor or silent partner.

In particular, columns (1) and (2) reveal that non-collateralized borrowing for educational purposes is positively and significantly associated with having SE business wealth in the next period. The probability increases by approximately 7.2 percentage points in the extensive margin model, while the magnitude of the intensive effect of a larger outstanding amount of the loan is comparatively smaller ($\beta = 0.008$). This finding is consistent with the role of human capital investment in facilitating entry into self-employment, as documented also in other studies (Block et al., 2013; Millán et al., 2014; Elert et al., 2015).

On the other hand, columns (3) and (4) show that, across all available financing instruments, borrowing to purchase a vehicle or other means of transport at time $t-1$ is associated with a higher probability of having non-SE business wealth at time t by approximately 1.3 percentage points. Again, the magnitude of a larger borrowed amount is economically smaller ($\beta = 0.001$), suggesting that access to credit plays a more important role than the amount borrowed in enabling households' investments in business activities. These results are consistent with other studies, where low-cost financing and vehicle ownership is associated with participation in the gig economy and higher employment prospects (Baum, 2009; Buchak, 2020; Van Doornik et al., 2024).

In addition, columns (5) and (6) show that households using the HMR as collateral to finance a business activity are more likely to receive income from non-SE business in the next period. The linear probability model indicates that the likelihood increases by approximately 15.6 percentage points, while the intensive margin effect is smaller in magnitude. This is consistent with households using borrowing to support additional income-generating activities other than their primary wealth source, and corroborates the hypothesis that confining household borrowing to consumption purposes only overshadows households' entrepreneurial scope and capacity.

Table 5 Collateral Channels, Loan Purpose, and Business Wealth Ownership

	DV: SE business wealth		DV: Non-SE business wealth		DV: Income from non-SE business	
	Purpose: Education		Purpose: Vehicle		Purpose: Business	
Financing channel	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level	(5) Dummy	(6) Debt Level
Non-collateralized loan$_{t-1}$	0.072**	0.008**				
	(0.033)	(0.003)				
All vehicle loans$_{t-1}$			0.013*	0.001*		
			(0.007)	(0.001)		
Main residence$_{t-1}$					0.156*	0.014*
					(0.085)	(0.007)
Observations	12,265	12,298	12,298	12,298	11,168	11,169
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Entries are coefficients from multiple-imputation linear probability models with household fixed effects. Columns are grouped by *loan purpose*. "Dummy" indicates the presence of the corresponding loan for the stated purpose (extensive margin). "Debt Level" is the inverse hyperbolic sine (IHS) transformation of the outstanding balance (intensive margin). All main regressors are lagged by one survey wave. Coefficients can be interpreted as changes in probability. Standard errors clustered at the household level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Finally, Table 6 evaluates whether these business loans have a positive effect on the value change of the non-SE business wealth. The findings reveal that, while collateralized borrowing may be more determinant in start-up capital, non-collateralized loans (NCLs) are associated with positive and statistically significant value changes. Specifically, having a NCL to finance a business in the previous period corresponds to approximately a 70% higher value change in non-SE business wealth in the next period. A 1% increase in the loan amount is associated with approximately a 7.1% increase in business value. These results are sizable, suggesting that such borrowing is associated with substantial increases in business asset values.

Table 6 Business and Renovation Borrowing and the Change in Non-SE Private Business Wealth

	DV: Δ IHS value of non-SE private business wealth			
	Purpose: Business		Purpose: Renovation	
Financing channel	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Non-collateralized loan$_{t-1}$	0.600*	0.071*	0.317*	0.036*
	(0.364)	(0.043)	(0.186)	(0.020)
Observations	12,265	12,298	12,265	12,298
Controls	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: Dependent variable is defined as the change in the IHS-transformed value of non-self-employment private business wealth between survey waves. Entries are coefficients from multiple-imputation fixed-effects (within) regressions with standard errors clustered at the household level in parentheses. Columns are grouped by *loan purpose*. "Dummy" indicates the presence of a non-collateralized loan for the stated purpose (extensive margin). "Debt Level" is the IHS-transformed outstanding balance for the stated purpose (intensive margin). All key regressors are lagged by one survey wave. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Moreover, the coefficients in columns (3) and (4) reveal that renovation loans are significantly linked to non-SE private business wealth. Specifically, the extensive margin effect of borrowing for renovation or refurbishing purposes increases the value change of the non-SE business by approximately 35%, while larger loans contribute approximately 3.6% for a 1% increase in the loan amount. This finding circles back to the results observed in Tables 3 and 4, and suggests that renovating properties used for business activities mainly serves supplementary wealth-generating ventures rather than primary businesses.

Taken together, these results show that the economic effects of household borrowing differ substantially across purposes, underscoring the importance of distinguishing debt by its intended use.

5.3 Rental Income

This section analyzes the hypothesis that housing purchases are related to rental income generation. The key explanatory variable is the lagged value of loans collateralized using the HMR to purchase or construct the HMR. As observed in Section 3.2, this is the most frequent loan–financing combination and represents the largest liability in household portfolios (Causa et al., 2019). Consequently, households may use housing assets to generate additional income flows, for instance by renting spare rooms or additional properties.

The coefficients in columns (1) and (2) report estimates from a linear probability model where the dependent variable is an indicator for receiving income from renting real estate at time t, upon borrowing to purchase the HMR at time $t-1$. The estimates are positive and statistically significant, and correspond to an increase of approximately 2.4 percentage points in probability. Similar to previous findings, the intensive margin effect indicates a smaller effect in magnitude ($\beta = 0.005$), suggesting that access to credit plays an important role for income-generating activities.

In addition, estimates of value changes in rental income are reported in columns (3) and (4). The extensive margin effect is associated with an approximately 41.5% increase in rental income, suggesting a strong relevance of the leveraged investment in pursuing additional income. By contrast, the intensive margin effect is negative and statistically significant. The result indicates that a 1% increase in the outstanding mortgage amount is associated with approximately an 8% decrease in rental income. This pattern may reflect heterogeneity in the reliance on rental income across the mortgage distribution. In fact, larger mortgages are likely associated with wealthier households, who may rely less on rental income as a source of financing to cushion the impact of the mortgage on their portfolios. This suggests that the relationship between borrowing and rental income is not monotonic, with credit access enabling rental activity while larger debt levels are associated with lower reliance on such income.

Table 7 Household Main Residence and Rental Income

	DV: Rental income		DV: Δ IHS value of rental income	
	Purpose: Buy HMR		Purpose: Buy HMR	
Financing channel	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Main residence$_{t-1}$	0.024*	0.005*	0.365**	-0.080*
	(0.013)	(0.003)	(0.183)	(0.045)
Observations	12,297	12,298	12,297	12,298
Controls	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: Columns (1)–(2) report coefficients from multiple-imputation linear probability models with household fixed effects, where the dependent variable is a binary indicator equal to one if the individual receives rental income. Columns (3)–(4) report coefficients from multiple-imputation fixed-effects regressions where the dependent variable is the change in the inverse hyperbolic sine (IHS) of rental income. "Dummy" indicates whether the household holds a collateralized loan to purchase the HMR. "Debt Level" denotes the IHS-transformed outstanding balance of the collateralized loan. All main regressors are lagged by one survey wave. Standard errors clustered at the household level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4 Employee Income

This last section tests the hypothesis that borrowing for educational purposes is associated with improved employment outcomes and higher future earnings. Table 8 shows the effect of education loans taken at time $t-1$ on employee income at time t, across all financing instruments available to households.

The results are positive and statistically significant in all specifications. In particular, having taken a loan to increase human capital in the previous period is associated with a 7 percentage point higher probability of having employee income in the next period. Albeit weaker in magnitude ($\beta = 0.006$), the intensive margin effect is also positive and statistically significant.

Table 8 Education Borrowing and Employee Income

	Employee income		Δ IHS value of employee income	
Education borrowing	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Has education loan$_{t-1}$	0.069**		1.102*	
	(0.031)		(0.572)	
Outstanding education debt$_{t-1}$		0.006**		0.113*
		(0.003)		(0.058)
Optimistic$_{t-1}$	-0.006	-0.006	0.049	0.049
	(0.009)	(0.009)	(0.156)	(0.156)
Observations	12,085	12,085	12,084	12,084
Controls	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: Columns (1)–(2) report coefficients from multiple-imputation linear probability models with household fixed effects, where the dependent variable is a binary indicator equal to one if the individual receives employee income. Columns (3)–(4) report coefficients from multiple-imputation fixed-effects regressions where the dependent variable is the change in the inverse hyperbolic sine (IHS) of employee income. "Dummy" indicates the presence of any education loan. "Debt Level" denotes the IHS-transformed outstanding balance of education loans. All main regressors are lagged by one survey wave. Coefficients in columns (1)–(2) can be interpreted as changes in probability. Standard errors clustered at the household level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

While more capable households may self-select into borrowing for education, the specification controls for future income expectations through the variable *Optimistic*. This variable captures households' expectations about future income growth, revealing their attitude toward future promotions and employment prospects, which may also affect their decision to borrow. However, its coefficient is small and statistically insignificant across all specifications, suggesting that expectations do not drive the main results.

By contrast, the extensive margin effect of the lagged value of education loans contributes significantly to the value change in employee income. Specifically, using the transformation proposed by Bellemare and Wichman (2020), having a loan for educational purposes corresponds to approximately a 155% increase in the value change of employee income. The significance of the effect is consistent with the role of education in improving labor market outcomes, as also shown in other studies analyzing the relationship (Avery and Turner, 2012; Black et al., 2023).

In addition, larger investments in education also show sizable earnings effects, implying that a 1% increase in the borrowed amount is associated with approximately an 11% increase in employee income. These results suggest that education-related borrowing is associated not only with higher employment likelihood, but also with substantial increases in earnings among employed households.

Overall, these findings underscore the importance of distinguishing household borrowing by its intended purpose, and offer relevant empirical evidence that favors a more nuanced approach to the analysis of household credit.

6 Conclusions

Before the global financial crisis of 2007-2008, the European securitization market was about three-quarters the size of U.S. issuance; by 2020, it had dropped to just 6%[4]. This difference reflects the profound impact that the crisis has had on the European lending market, which also transpires in the post-crisis academic literature. In fact, while much of the existing literature analyzes the macroeconomic consequences associated with excessive household debt, less attention has been paid to how households allocate their borrowing across functions and financing instruments. This analysis is particularly relevant at a time when most European countries are looking for ways to inject fresh cash into their struggling economies, with European policy makers debating how to incentivize households to invest their substantial savings into productive activities. This paper argues that one step in that direction is to revisit the traditional association between household borrowing and consumption spending, providing novel evidence regarding the motivations behind household leverage.

Using panel data from the HFCS across five European countries, this paper examines how purpose-specific borrowing relates to business and income outcomes. The analysis relies on fixed-effects models and clustered standard errors that exploit within-household variation over time.

The results provide consistent evidence in support of the proposed hypotheses. In particular, the findings reveal that households collateralize real estate other than the HMR to purchase additional properties destined to business activities. Moreover, loans taken to renovate or refurbish the property are associated with both a higher likelihood of holding real estate for business and with an increase in its value. In addition, the renovated properties contribute to household non-self employment (SE) wealth, consistent with households using renovation loans to generate additional revenue beyond the primary wealth source.

Similarly, borrowing to finance a business or a professional activity corresponds to both a higher likelihood of receiving non-SE income and an increase in non-SE business wealth. This reinforces the relevant role of borrowing as an external source of funding used by households to generate additional revenue streams rather than for

[4]European Stability Mechanism: Reviving securitization in Europe for CMU. https://www.esm.europa.eu/blog/reviving-securitisation-europe-cmu. Accessed 10 March 2025

living expenses or similar purchases. Likewise, loans to purchase a vehicle or other means of transport are associated with a higher probability of having non-SE business wealth, suggesting that owning a vehicle reduces spatial constraints and increases business opportunities (Baum, 2009; Buchak, 2020; Van Doornik et al., 2024). By contrast, borrowing for educational purposes is linked with a higher probability of having SE business wealth, consistent with the role of human capital development in spurring entry into self-employment (Block et al., 2013; Millán et al., 2014; Elert et al., 2015). In addition, leveraged investments in education are associated with both a higher probability of receiving employee income and with substantial increases in future earnings, even after controlling for initial wealth levels, employment status, and future income expectations.

Finally, collateralized investments in housing are associated with a higher likelihood of receiving rental income from real estate, indicating that households look to the rental market as an additional income source to offset the impact of what for many is their largest liability (Causa et al., 2019). However, this pattern does not hold along the entire mortgage distribution, as larger outstanding mortgage amounts are negatively associated with value changes in rental income. Since wealthier households are more likely to obtain access to more credit, the finding reveals less reliance on rental income across the mortgage distribution.

A key finding consistent across models and specifications is that the extensive margin is more strongly associated with the corresponding outcomes than the intensive margin. This heterogeneity in magnitude reflects the relevance that credit availability plays in households' ability to create additional wealth, rather than the actual amount borrowed, and suggests that tight provisions of start-up capital may excessively limit households' financial growth.

Overall, the findings highlight that the economic effects of household borrowing depend critically on how credit is used, rather than on its aggregate level alone. While much of the existing literature emphasizes the risks deriving from excessive leverage, this paper documents several channels through which borrowing is associated with wealth accumulation and income generation. Turning the spotlight onto these mechanisms is especially relevant today, as European economies look for investments to revitalize a market union battered by many crises. As stringent financial regulations and the limited availability of resources constrain governments' ability to expand public spending, the results suggest that facilitating access to credit for productive uses may support household-level financial development and broader economic activity, while preserving safeguards against excessive risk-taking.

Declarations

- Funding: This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.
- Competing interests: The author has no competing interests to declare that are relevant to the content of this article.
- Ethics approval and consent to participate: 'Not applicable'.
- Consent for publication: 'Not applicable'.
- Data availability: The dataset analyzed during the current study is available upon request to researchers at https://www.ecb.europa.eu/stats/ecb_surveys/hfcs/html/index.en.html.
- Materials and code availability: All materials and code used in the analysis are available upon request to the author.
- Author contribution: The author confirms the sole responsibility for the conception of the study, performed analysis and manuscript preparation.

Appendix A Supplementary Tables

Table A.1 HFCS variable definition: outcomes and controls

Analysis variable	Survey question / definition	Coding
Business outcomes		
Has other real estate property for business	Indicator for whether the household holds any (other) properties used for business activities, such as houses, apartments, garages, offices, hotels, other commercial buildings, farms, land, etc. Includes properties in the survey country and abroad.	1=Yes, 0=No
Value of other real estate property for business activities	"What is the value of this property, i.e. if you could sell it now how much do you think would be the price of the property?"	Numeric (EUR)
Has self-employment business wealth	"(Do you/Does anyone in your household) own all or part of any business that is not publicly traded where you have an active role in running the business?"	1=Yes, 0=No
Has income from private business other than self-employment	"(Other than self-employment income I have already recorded, did/Did) (you/your household) receive any income from a private business or partnership in (the last 12 months / the last calendar year)?"	1=Yes, 0=No
Has non-SE private business wealth	"(Do you/Does anyone in your household) own any business that is not publicly traded where you/they act only as an investor or silent partner?"	1=Yes, 0=No
Value of non self-employment not publicly traded businesses	"What is the value of your (your household's) share of (this business/these businesses)?"	Numeric (EUR)
Earnings outcomes		
Received rental income	"Did you/your household receive any income from renting real estate in the last 12 months / the last calendar year?"	1=Yes, 2=No
Gross rental income	"What was the total gross amount over the last 12 months / the last calendar year?"	Numeric (EUR)
Received employee income	"Did you receive any sort of employee income during (last 12 months / last calendar year)?"	1=Yes, 2=No
Gross cash employee income	"What was the total gross amount over (the last 12 months / last calendar year)?" (regular wages/salaries, overtime, tips, bonuses, profit sharing benefits unless part of pension arrangements).	Numeric (EUR)
Income and baseline wealth controls		
Gross household income, equivalised	Total gross annual household income aggregate, equalised using the modified OECD equivalence scale.	Numeric (EUR)
Future income expectations	"Over the next year, do you expect your (household's) total income to go up more than prices, less than prices, or about the same as prices?"	1=More than prices, 2=Less than prices, 3=About the same as prices
Wealth quintile (1–5)	Household position in the national net wealth distribution, computed within country.	1–5
Demographic controls (reference person, Canberra definition)		
Household size	Number of household members (all household members included).	Numeric
Age of reference person	Age of the household reference person.	Numeric (years)
Marital status	"What is X's (your) marital status?"	Ordered categories[a]
Labour status	Main labour status of the reference person.	Ordered categories[b]
Gender	Gender of the reference person.	0=Male, 1=Female
Education	Highest educational attainment of the reference person (Canberra definition).	Ordered categories[c]

Notes: The household reference person follows the UN/Canberra definition.
[a] Marital status coding: 1=Single, 2=Married, 3=Consensual union, 4=Widowed, 5=Divorced.
[b] Labour status coding: 1=Employee, 2=Self-employed, 3=Unemployed, 4=Retired, 5=Other.
[c] Education coding: 0=Early childhood/no education; 1=Primary or no formal education; 2=Lower secondary; 3=Upper secondary; 4=Post-secondary non-tertiary; 5=Short-cycle tertiary; 6=Bachelor or equivalent.

Table A.2 HFCS variable definition: financing and purposes

Analysis variable	Survey question / definition	Coding
Financing and purposes		
HMR collateralised mortgages/loans	"Are there currently any outstanding mortgages or loans that use the residence as collateral?"	1=Yes, 2=No
HMR mortgage: purpose of the loan	When first taking out this mortgage, what was the purpose for which the money was used? Set of up to four items (a–d), with (a) main purpose.	Purpose categories[a]
HMR mortgage: current amount outstanding	"What is the amount still owed on the loan?" (outstanding principal excluding interest/fees).	Numeric (EUR)
Other property: collateralised mortgages/loans	"Are there currently any outstanding mortgages or loans that use this property as collateral?"	1=Yes, 2=No
Other property mortgage: purpose of the loan	When first taking out this mortgage, what was the purpose for which the money was used? Set of up to four items (a–d), with (a) main purpose.	Purpose categories[a]
Other property mortgage: current amount outstanding	"What is the amount still owed on the loan?" (outstanding principal excluding interest/fees).	Numeric (EUR)
Has any non-collateralised loans	"(Other than loans I have already recorded), do you have any (other) loans or owe any (other) money (e.g. car loans, consumer loans, instalment loans, employer loans, etc.)?"	1=Yes, 2=No
Non-collateralised loan: purpose of the loan	Why did you take on this loan? Set of up to four items (a–d), with (a) main purpose.	Purpose categories[b]
Non-collateralised loan: outstanding balance	"What is the outstanding balance on the loan?"	Numeric (EUR)
Has private loans	"(Other than loans I have already recorded), do you have loans from relatives or friends that you are expected to repay?"	1=Yes, 2=No
Private loan: purpose of the loan	Why did you take on this loan? Set of up to four items (a–d), with (a) main purpose.	Purpose categories[a]
Private loan: outstanding amount	"How much is the (total) outstanding balance?/How much are you still expected to repay?"	Numeric (EUR)

Notes: Each loan instrument records up to four purposes (a–d); (a) is the main purpose.
[a] Purpose categories (HMR mortgages; other-property mortgages; private loans): (1) purchase/construct (HMR / this property); (2) purchase other real estate; (3) refurbish/renovate property; (4) vehicle/transport; (5) finance business/professional activity; (6) consolidate/refinance debts; (7) education; (8) living expenses/other purchases; (9) other.
[b] Purpose categories (non-collateralised loans): (1) purchase/construct the HMR; (2) purchase other real estate; (3) refurbish/renovate residence; (4) vehicle/transport; (5) finance business/professional activity; (6) consolidate debts; (7) education; (8) living expenses/other purchases; (9) other; (10) support relatives and friends.

Appendix B Regressions with Controls

Table B.1 Collateral Channels, Loan Purpose, and the Probability of Holding Real Estate for Business

	DV: Holding real estate for business			
	Purpose: Buy other properties		Purpose: Renovation	
	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Financing channel				
Other properties$_{t-1}$	0.033**	0.003**		
	(0.015)	(0.001)		
Main residence$_{t-1}$			0.026*	0.002*
			(0.015)	(0.001)
Controls				
Wealth quintile	0.012***	0.012***	0.012***	0.012***
	(0.003)	(0.003)	(0.003)	(0.003)
Equivalised income	0.002	0.002	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
Household members	0.002	0.002	0.003	0.003
	(0.002)	(0.002)	(0.002)	(0.002)
Age	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Tertiary education	-0.002	-0.002	-0.002	-0.002
	(0.007)	(0.007)	(0.007)	(0.007)
Employed	0.025***	0.026***	0.026***	0.026***
	(0.008)	(0.008)	(0.008)	(0.008)
Female	-0.006	-0.006	-0.006	-0.006
	(0.007)	(0.007)	(0.007)	(0.007)
Married	0.007	0.007	0.008	0.008
	(0.007)	(0.007)	(0.007)	(0.007)
Year dummies				
2017	0.005	0.005	0.004	0.004
	(0.004)	(0.004)	(0.004)	(0.004)
2020	0.004	0.003	0.003	0.003
	(0.004)	(0.004)	(0.004)	(0.004)
Observations	12,286	12,298	12,297	12,298
Household FE	Yes	Yes	Yes	Yes

Notes: Entries are coefficients from multiple-imputation linear probability models with household fixed effects. The dependent variable is a binary indicator equal to one if the household holds real estate for business purposes. Columns are grouped by the *purpose of the collateralized loan*. "Dummy" indicates whether the household holds a collateralized loan for the stated purpose (extensive margin). "Debt Level" is the inverse hyperbolic sine (IHS) transformation of the outstanding balance of the collateralized loan for the stated purpose (intensive margin). All main regressors are lagged by one survey wave. Coefficients can be interpreted as changes in probability. Standard errors clustered at the household level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2 Renovation Borrowing and the Change in the Value of Real Estate for Business

	DV: Δ IHS value of real estate for business			
	Purpose: Renovation			
	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Financing channel				
Main residence$_{t-1}$	0.470**	0.041**		
	(0.232)	(0.021)		
Non-collateralized loan$_{t-1}$			0.439*	0.050**
			(0.244)	(0.024)
Controls				
Wealth quintile	0.205***	0.205***	0.208***	0.207***
	(0.055)	(0.055)	(0.056)	(0.055)
Equivalised income	0.039	0.039	0.040	0.038
	(0.040)	(0.040)	(0.041)	(0.040)
Household members	0.036	0.036	0.037	0.038
	(0.080)	(0.080)	(0.081)	(0.080)
Age	0.008	0.008	0.008	0.008
	(0.009)	(0.009)	(0.009)	(0.009)
Tertiary education	-0.065	-0.065	-0.059	-0.060
	(0.132)	(0.132)	(0.132)	(0.132)
Employed	0.263*	0.263*	0.256	0.264*
	(0.157)	(0.156)	(0.157)	(0.156)
Female	-0.089	-0.089	-0.077	-0.088
	(0.136)	(0.136)	(0.135)	(0.136)
Married	0.049	0.049	0.035	0.042
	(0.140)	(0.140)	(0.141)	(0.140)
Year dummies				
2017	0.120	0.120	0.127	0.125
	(0.078)	(0.078)	(0.078)	(0.078)
2020	0.056	0.056	0.053	0.058
	(0.076)	(0.076)	(0.077)	(0.076)
Observations	12,297	12,298	12,265	12,298
Household FE	Yes	Yes	Yes	Yes

Notes: Dependent variable is defined as the change in the inverse hyperbolic sine (IHS) value of other real estate property used for business activities between survey waves. Entries are coefficients from multiple-imputation fixed-effects (within) regressions with standard errors clustered at the household level in parentheses. "Dummy" indicates the presence of a renovation loan (extensive margin). "Debt (IHS)" is the IHS-transformed outstanding renovation debt (intensive margin). All renovation borrowing variables are lagged by one survey wave. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.3 Collateral Channels, Loan Purpose, and Business Wealth Ownership

	DV: SE business wealth		DV: Non-SE business wealth		DV: Income from non-SE business	
	Purpose: Education		Purpose: Vehicle		Purpose: Business	
Financing channel	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level	(5) Dummy	(6) Debt Level
Non-collateralized loan$_{t-1}$	0.072**	0.008**				
	(0.033)	(0.003)				
All vehicle loans$_{t-1}$			0.013*	0.001*		
			(0.007)	(0.001)		
Main residence$_{t-1}$					0.156*	0.014*
					(0.085)	(0.007)
Controls						
Household size	0.033***	0.033***	-0.002	-0.002	0.000	0.000
	(0.006)	(0.006)	(0.003)	(0.003)	(0.004)	(0.004)
Wealth quintile	0.038***	0.037***	0.003	0.003	0.011***	0.011***
	(0.005)	(0.005)	(0.002)	(0.002)	(0.003)	(0.003)
Equivalised income	0.004	0.004	0.000	0.000	–	–
	(0.003)	(0.003)	(0.002)	(0.002)		
Age	0.001*	0.001*	0.000	0.000	0.000	0.000
	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Tertiary education	-0.027**	-0.027**	0.014**	0.014**	-0.003	-0.003
	(0.012)	(0.012)	(0.007)	(0.007)	(0.008)	(0.008)
Employed	0.121***	0.122***	-0.006	-0.006	0.007	0.007
	(0.013)	(0.013)	(0.006)	(0.006)	(0.007)	(0.007)
Female	-0.012	-0.012	-0.008	-0.008	0.010	0.010
	(0.011)	(0.011)	(0.005)	(0.005)	(0.007)	(0.007)
Married	-0.005	-0.005	-0.003	-0.003	0.007	0.007
	(0.014)	(0.014)	(0.007)	(0.007)	(0.008)	(0.008)
Year dummies						
2017	-0.014***	-0.014***	0.009***	0.009***	0.012***	0.012***
	(0.005)	(0.005)	(0.003)	(0.003)	(0.004)	(0.004)
2020	-0.019***	-0.020***	0.003	0.003	0.016***	0.016***
	(0.007)	(0.007)	(0.003)	(0.003)	(0.004)	(0.004)
Observations	12,265	12,298	12,298	12,298	11,168	11,169
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Entries are coefficients from multiple-imputation linear probability models with household fixed effects. Columns are grouped by *loan purpose*. "Dummy" indicates the presence of the corresponding loan for the stated purpose (extensive margin). "Debt Level" is the inverse hyperbolic sine (IHS) transformation of the outstanding balance (intensive margin). All main regressors are lagged by one survey wave. Coefficients can be interpreted as changes in probability. Standard errors clustered at the household level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.4 Business and Renovation Borrowing and the Change in Non-SE Private Business Wealth

	DV: Δ IHS value of non-SE private business wealth			
	Purpose: Business		Purpose: Renovation	
	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Financing channel				
Non-collateralized loan$_{t-1}$	0.600*	0.071*	0.317*	0.036*
	(0.364)	(0.043)	(0.186)	(0.020)
Controls				
Wealth quintile	-0.021	-0.021	-0.021	-0.021
	(0.048)	(0.048)	(0.048)	(0.048)
Equivalised income	0.016	0.016	0.016	0.016
	(0.039)	(0.039)	(0.039)	(0.039)
Household members	-0.021	-0.020	-0.021	-0.021
	(0.048)	(0.048)	(0.048)	(0.047)
Age	0.002	0.002	0.001	0.001
	(0.006)	(0.006)	(0.006)	(0.006)
Tertiary education	0.213**	0.212**	0.214**	0.214**
	(0.107)	(0.107)	(0.107)	(0.107)
Employed	-0.078	-0.078	-0.075	-0.074
	(0.131)	(0.130)	(0.131)	(0.130)
Female	-0.196*	-0.195*	-0.191*	-0.190*
	(0.102)	(0.102)	(0.102)	(0.101)
Married	-0.029	-0.030	-0.032	-0.031
	(0.117)	(0.117)	(0.117)	(0.117)
Year dummies				
2017	0.140**	0.139**	0.138**	0.137**
	(0.057)	(0.057)	(0.057)	(0.057)
2020	-0.022	-0.022	-0.024	-0.024
	(0.058)	(0.057)	(0.058)	(0.057)
Observations	12,265	12,298	12,265	12,298
Household FE	Yes	Yes	Yes	Yes

Notes: Dependent variable is defined as the change in the IHS-transformed value of non-self-employment private business wealth between survey waves. Entries are coefficients from multiple-imputation fixed-effects (within) regressions with standard errors clustered at the household level in parentheses. Columns are grouped by loan purpose. "Dummy" indicates the presence of a non-collateralized loan for the stated purpose (extensive margin). "Debt (IHS)" is the IHS-transformed outstanding balance for the stated purpose (intensive margin). All key regressors are lagged by one survey wave.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.5 Household Main Residence and Rental Income

	DV: Rental income		DV: Δ IHS value of rental income	
	Purpose: Buy HMR		Purpose: Buy HMR	
	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Financing channel				
Main residence$_{t-1}$	0.025*	0.005*	0.365**	-0.081*
	(0.013)	(0.003)	(0.183)	(0.045)
Controls				
Household members	0.003	0.003	0.046	0.056
	(0.005)	(0.005)	(0.068)	(0.068)
Wealth quintile	0.039***	0.039***	0.449***	0.452***
	(0.005)	(0.005)	(0.067)	(0.067)
Age	0.002***	0.002***	0.027***	0.028***
	(0.001)	(0.001)	(0.010)	(0.010)
Tertiary	0.016	0.016	0.044	0.047
	(0.014)	(0.014)	(0.195)	(0.195)
Employed	0.019*	0.019*	0.361**	0.374***
	(0.010)	(0.010)	(0.140)	(0.141)
Female	-0.002	-0.002	-0.099	-0.099
	(0.011)	(0.011)	(0.151)	(0.150)
Married	0.033**	0.034**	0.120	0.104
	(0.015)	(0.016)	(0.208)	(0.207)
Year dummies				
2017	0.012**	0.012**	-0.002	-0.022
	(0.005)	(0.005)	(0.078)	(0.078)
2020	0.005	0.004	-0.211**	-0.232***
	(0.007)	(0.007)	(0.088)	(0.087)
Observations	12,297	12,298	12,297	12,298
Household FE	Yes	Yes	Yes	Yes

Notes: Columns (1)–(2) report coefficients from multiple-imputation linear probability models with household fixed effects, where the dependent variable is a binary indicator equal to one if the individual receives rental income. Columns (3)–(4) report coefficients from multiple-imputation fixed-effects regressions where the dependent variable is the change in the inverse hyperbolic sine (IHS) of rental income. "Dummy" indicates whether the household holds a collateralized loan to purchase the HMR. "Debt Level" denotes the IHS-transformed outstanding balance of the collateralized loan. All main regressors are lagged by one survey wave. Standard errors clustered at the household level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.6 Education Borrowing and Employee Income

	Employee income		Δ IHS value of employee income	
	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Education borrowing				
Has education loan$_{t-1}$	0.069**		1.102*	
	(0.031)		(0.572)	
Outstanding education debt$_{t-1}$		0.007**		0.113*
		(0.003)		(0.058)
Controls				
Optimistic$_{t-1}$	-0.006	-0.006	0.049	0.049
	(0.009)	(0.009)	(0.156)	(0.156)
Household members	0.071***	0.071***	1.147***	1.148***
	(0.007)	(0.007)	(0.096)	(0.096)
Wealth quintile	0.006	0.006	-0.108	-0.108
	(0.005)	(0.005)	(0.080)	(0.080)
Age	-0.004***	-0.004***	-0.033**	-0.033**
	(0.001)	(0.001)	(0.013)	(0.013)
Tertiary education	0.015	0.015	0.144	0.145
	(0.014)	(0.014)	(0.255)	(0.255)
Employed	0.330***	0.330***	4.010***	4.009***
	(0.018)	(0.018)	(0.244)	(0.244)
Female	0.005	0.005	0.183	0.183
	(0.012)	(0.012)	(0.195)	(0.195)
Married	0.013	0.012	-0.203	-0.203
	(0.017)	(0.017)	(0.264)	(0.264)
Year dummies				
2017	-0.009	-0.009	0.611***	0.610***
	(0.006)	(0.006)	(0.102)	(0.102)
2020	-0.009	-0.009	1.048***	1.047***
	(0.008)	(0.008)	(0.118)	(0.118)
Observations	12,085	12,085	12,084	12,084
Household FE	Yes	Yes	Yes	Yes

Notes: Columns (1)–(2) report coefficients from multiple-imputation linear probability models with household fixed effects, where the dependent variable is a binary indicator equal to one if the individual receives employee income. Columns (3)–(4) report coefficients from multiple-imputation fixed-effects regressions where the dependent variable is the change in the inverse hyperbolic sine (IHS) of employee income. “Dummy” indicates the presence of any education loan. “Debt Level” denotes the IHS-transformed outstanding balance of education loans. All main regressors are lagged by one survey wave. Coefficients in columns (1)–(2) can be interpreted as changes in probability. Standard errors clustered at the household level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix C Conditional Fixed-Effects Logit Estimates

Table C.1 Collateral Channels, Loan Purpose, and the Odds of Holding Real Estate for Business

	DV: Holding real estate for business			
	Purpose: Buy other properties		Purpose: Renovation	
	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Financing channel				
Other properties$_{t-1}$	1.982**	1.054*		
	(0.649)	(0.028)		
Main residence$_{t-1}$			5.179**	1.159**
			(3.826)	(0.075)
Controls				
Wealth quintile	1.901***	1.884***	1.903***	1.907***
	(0.314)	(0.309)	(0.303)	(0.303)
Equivalised income	1.037	1.036	1.029	1.029
	(0.044)	(0.045)	(0.045)	(0.045)
Household members	1.015	1.025	1.025	1.026
	(0.136)	(0.136)	(0.139)	(0.139)
Age	1.005	1.010	1.012	1.012
	(0.019)	(0.019)	(0.019)	(0.019)
Tertiary education	1.009	1.017	0.982	0.983
	(0.379)	(0.382)	(0.383)	(0.381)
Employed	2.314**	2.320***	2.424***	2.421***
	(0.749)	(0.719)	(0.758)	(0.757)
Female	0.912	0.936	0.910	0.910
	(0.281)	(0.281)	(0.273)	(0.273)
Married	2.128	2.168	2.236	2.220
	(1.259)	(1.296)	(1.336)	(1.322)
Year dummies				
2017	1.277	1.286	1.266	1.268
	(0.210)	(0.211)	(0.204)	(0.205)
2020	1.347	1.308	1.287	1.286
	(0.284)	(0.271)	(0.266)	(0.266)
Observations	829	837	837	837
Household FE	Yes	Yes	Yes	Yes

Notes: Entries are odds ratios from multiple-imputation conditional fixed-effects logit models. Columns are grouped by the *purpose of the collateralized loan*. "Dummy" indicates whether the household holds a collateralized loan for the stated purpose (extensive margin). "Debt Level" is the IHS-transformed outstanding balance of the collateralized loan for the stated purpose (intensive margin). "Other properties" refers to loans collateralized by other real estate; "Main residence" refers to loans collateralized by the household main residence. All main regressors are lagged by one survey wave. Standard errors clustered at the household level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.2 Collateral Channels, Loan Purpose, and Business Wealth Ownership

	DV: SE business wealth		DV: Non-SE business wealth		DV: Income from non-SE business	
	Purpose: Education		Purpose: Vehicle		Purpose: Business	
	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level	(5) Dummy	(6) Debt Level
Financing channel						
Non-collateralized loan$_{t-1}$	5.663	1.218				
	(7.337)	(0.183)				
Vehicle loan$_{t-1}$			2.394*	1.094*		
			(1.236)	(0.057)		
Main residence$_{t-1}$					8.008**	1.190**
					(7.936)	(0.093)
Controls						
Household size	1.453***	1.447***	0.846	0.846	0.948	0.947
	(0.157)	(0.155)	(0.159)	(0.159)	(0.142)	(0.142)
Wealth quintile	2.445***	2.374***	1.259	1.255	1.845**	1.852**
	(0.329)	(0.312)	(0.246)	(0.245)	(0.425)	(0.430)
Equivalised income	1.067	1.075	1.009	1.009	–	–
	(0.043)	(0.049)	(0.031)	(0.031)		
Age	0.989	0.990	0.993	0.993	1.010	1.010
	(0.015)	(0.015)	(0.024)	(0.024)	(0.022)	(0.022)
Tertiary education	0.593*	0.603*	2.225**	2.221**	0.843	0.853
	(0.163)	(0.165)	(0.879)	(0.877)	(0.344)	(0.349)
Employed	5.389***	5.367***	0.717	0.713	1.843	1.364
	(1.202)	(1.195)	(0.244)	(0.242)	(0.344)	(0.496)
Female	0.804	0.795	0.539	0.539	1.317	1.322
	(0.167)	(0.164)	(0.235)	(0.236)	(0.464)	(0.465)
Married	1.294	1.281	1.004	1.008	1.611	1.627
	(0.437)	(0.431)	(0.580)	(0.581)	(1.030)	(1.041)
Year dummies						
2017	0.811*	0.800*	1.728***	1.733***	1.710**	1.702**
	(0.098)	(0.096)	(0.333)	(0.334)	(0.371)	(0.370)
2020	0.717**	0.702**	1.258	1.258	2.373***	2.361***
	(0.111)	(0.109)	(0.307)	(0.307)	(0.658)	(0.655)
Observations	1,700	1,710	636	636	559	559
Household FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Entries are odds ratios from multiple-imputation conditional fixed-effects logit models. Columns are grouped by *loan purpose*. "Dummy" indicates the presence of the corresponding loan for the stated purpose (extensive margin). "Debt Level" is the inverse hyperbolic sine (IHS) transformation of the outstanding balance (intensive margin). All main regressors are lagged by one survey wave. Standard errors clustered at the household level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.3 Collateralized Borrowing and Income Outcomes

	DV: Rental income		DV: Employee income	
	Purpose: Buy HMR		Purpose: Education	
	(1) Dummy	(2) Debt Level	(3) Dummy	(4) Debt Level
Financing channel				
Main residence$_{t-1}$	1.679**	1.060**		
	(0.366)	(0.027)		
All education loans$_{t-1}$			5.626**	1.168**
			(4.148)	(0.073)
Controls				
Wealth quintile	2.100***	2.097***	1.044	1.044
	(0.208)	(0.202)	(0.090)	(0.091)
Household members	1.091	1.072	2.941***	2.936***
	(0.118)	(0.112)	(0.398)	(0.340)
Age	1.026*	1.027**	0.956***	0.956***
	(0.014)	(0.013)	(0.013)	(0.013)
Tertiary education	1.368	1.379	1.166	1.159
	(0.328)	(0.304)	(0.299)	(0.296)
Employed	1.253	1.250	10.038***	9.999***
	(0.259)	(0.260)	(1.856)	(1.763)
Female	1.000	0.999	1.158	1.159
	(0.188)	(0.183)	(0.228)	(0.220)
Married	1.462	1.516	1.658	1.636
	(0.408)	(0.389)	(0.624)	(0.547)
Year dummies				
2017	1.348***	1.360***	0.858	0.859
	(0.141)	(0.150)	(0.092)	(0.096)
2020	1.198	1.178	0.772*	0.772*
	(0.163)	(0.153)	(0.111)	(0.105)
Observations	1,896	1,896	2,685	2,685
Household FE	Yes	Yes	Yes	Yes

Notes: Entries are odds ratios from multiple-imputation conditional fixed-effects logit models. Columns (1)–(2) examine the relationship between borrowing for main residence purchase and the probability of receiving rental income. Columns (3)–(4) examine the relationship between education borrowing and the probability of receiving employee income. “Dummy” indicates the presence of the corresponding loan (extensive margin). “Debt Level” denotes the inverse hyperbolic sine (IHS) of the outstanding balance (intensive margin). All regressors are lagged by one survey wave. Standard errors clustered at the household level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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